

- 3 The photograph shows a Javanese edelweiss plant.



(Source: © Ciptor / Shutterstock)

- (a) This plant is endemic to Indonesia.

State what is meant by the term **endemic**.

(1)

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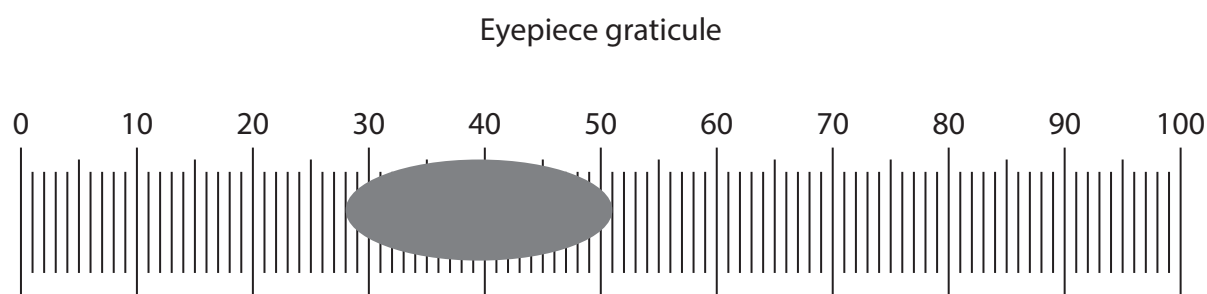
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- (b) A student used a microscope and a graticule to view a pollen grain from this plant.



One hundred eyepiece graticule divisions is 0.1 mm.

- (i) Which is the length of this pollen grain?

(1)

- ☐ A 2.3 mm
- ☐ B 23 μm
- ☐ C 230 μm
- ☐ D 2300 nm

- (ii) State what is meant by the term **magnification**.

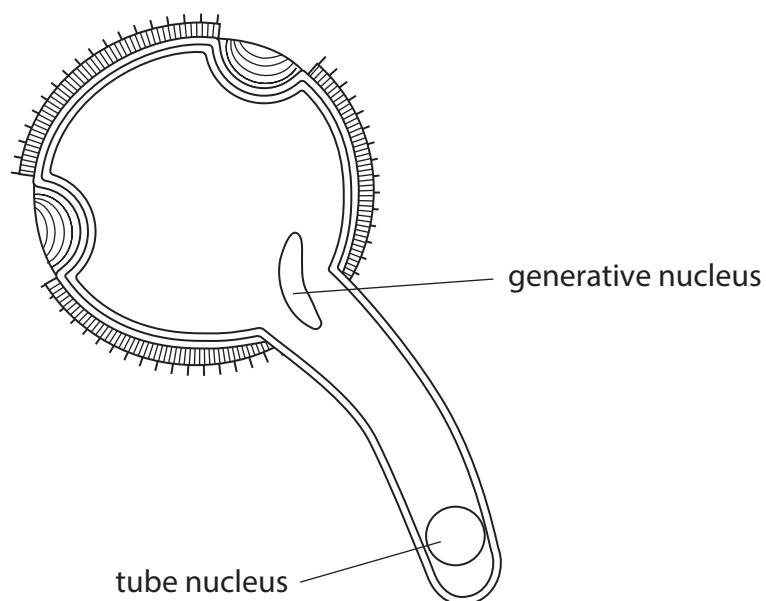
(1)

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(c) The diagram shows a germinating pollen grain.



Describe the roles of the tube nucleus and the generative nucleus.

(4)

Tube nucleus

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Generative nucleus

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(Total for Question 3 = 7 marks)